

TELCO CHURN PREDICTOR & RETENTION ENGINE

FROM RAW DATA TO A
BUSINESS-READY CLOUD
APPLICATION.

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ROBUST PIPELINE ARCHITECTURE

DATA INGESTION

Raw data reading, null handling, and initial formatting.



TRANSFORMATION (OOP)

Custom Object-Oriented classes (TransformerMixin) for safe scaling and encoding.



SECURE MODELING

Implementation of Nested Cross-Validation to completely prevent Data Leakage.

STATISTICAL RIGOR: MODEL & STRATEGY

Comparable performance across all top models and strategies. Granular Logistic Regression was selected for its high explainability and low latency.

Strategy	Model	F1 Mean	AUC Mean
Granular	LightGBM	0.6284	0.8421
Granular	XGBoost	0.6277	0.8456
Granular	Random Forest	0.6269	0.8420
Granular	Logistic Regression	0.6268	0.8431
Aggregated	LightGBM	0.6262	0.8402
Aggregated	Random Forest	0.6259	0.841
Aggregated	Logistic Regression	0.6256	0.8412
Aggregated	XGBoost	0.6244	0.8423

PRODUCTION PIPELINE: FINAL HPO & TRAINING ON 100% OF DATA • DECISION THRESHOLD MATHEMATICALLY OPTIMIZED (**YOUDEN'S J = 0.48**) TO MAXIMIZE ACTUAL CHURN DETECTION.

FROM NOTEBOOK TO PRODUCTION

Interactive web application deployed on Streamlit Community Cloud.

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 **Telco Churn Predictor & Retention Engine**

This application predicts the likelihood of a customer leaving telecom service and automatically generates corporate retention tactics for the sales team.

 **Client Profile**

Gender

Female

Is Senior Citizen? (0 = NO, 1 = YES)

0

Has Partner?

Yes

Has Dependents?

Yes

Tenure (Months)

12

 **Internet Services**

Internet Service

DSL

Multiple Lines

No

Online Security

No

Online Backup

No

Device Protection

No

Tech Support

No

Streaming TV

No

Streaming Movies

No

 **Contract and Payments**

Contract Type

Month-to-month

Paperless Billing

Yes

Payment Method

Electronic check

Monthly Charge (\$)

50.00

< Manage app

BUSINESS IMPACT: PREDICTION & RETENTION

Tenure (Months)

12

Device Protection

No

Tech Support

No

Streaming TV

No

Streaming Movies

No

Evaluate Churn Risk & Generate Action Plan

Analysis Results

HIGH CHURN RISK (60.8%)

The customer exceeds the critical risk threshold of 48%. Immediate preventive action is strongly advised.

Recommended Next Best Actions (NBA)

Based on this customer's features, apply the following customized commercial triggers:

- Contract Upgrade:** The customer has no binding contract. Offer a 15% discount for the next 6 months in exchange for upgrading to a stable 1-Year contract today.
- Value-Added Loyalty Tactic:** Provide 3 months of Premium Tech Support for free. Historical telecom data proves that customers with active technical assistance have significantly higher retention rates.

Predicting churn is only half the battle.

The system automatically prescribes customized Next Best Actions (NBA) for the sales team based on each customer's unique profile.

Examples: Targeted contract upgrades (e.g., 15% discount) or value-added loyalty tactics (e.g., free premium tech support).